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Decoding continuous sleep states in held-out mice from regional fMRI dynamics

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INTRODUCTION

Human fMRI studies have validated sleep-stage classification across subjects[1], while mouse fMRI studies have predicted state transitions[2]. We tested whether regional fMRI dynamics could reconstruct continuous sleep states in mice, and assessed temporal decoding and regional contributions.

METHODS

We analyzed 45 sessions from 24 mice (180 hours), with electrophysiological Awake, NREM and REM labels.

From 54 anatomical ROIs, we extracted five features per ROI to obtain 270 inputs for elastic-net multinomial logistic regression. The selected centered windows spanned 21 TR (TR = 2 s).

Nested leave-one-mouse-out validation separated training, tuning and testing. An HMM then decoded full sessions offline.

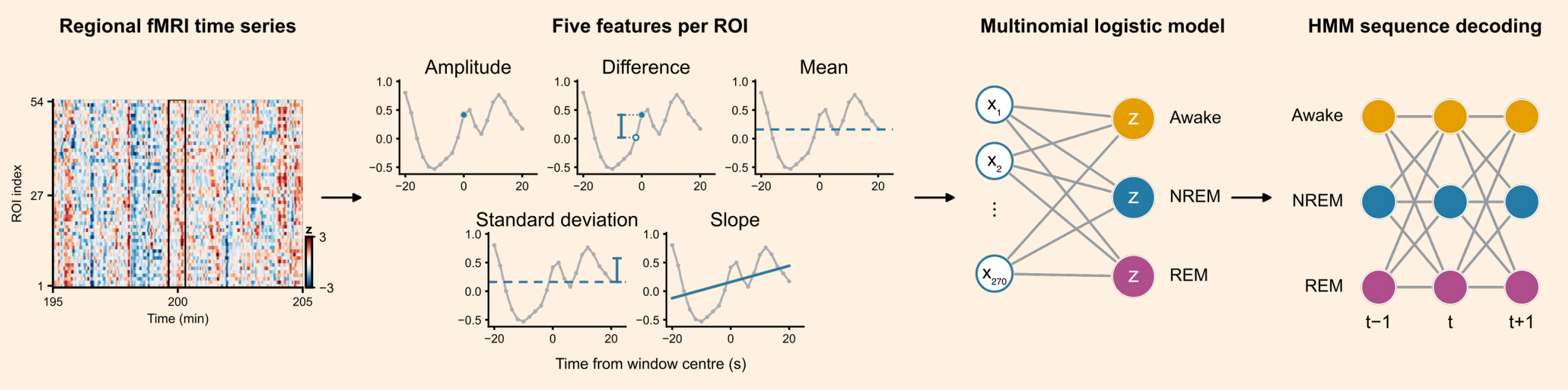
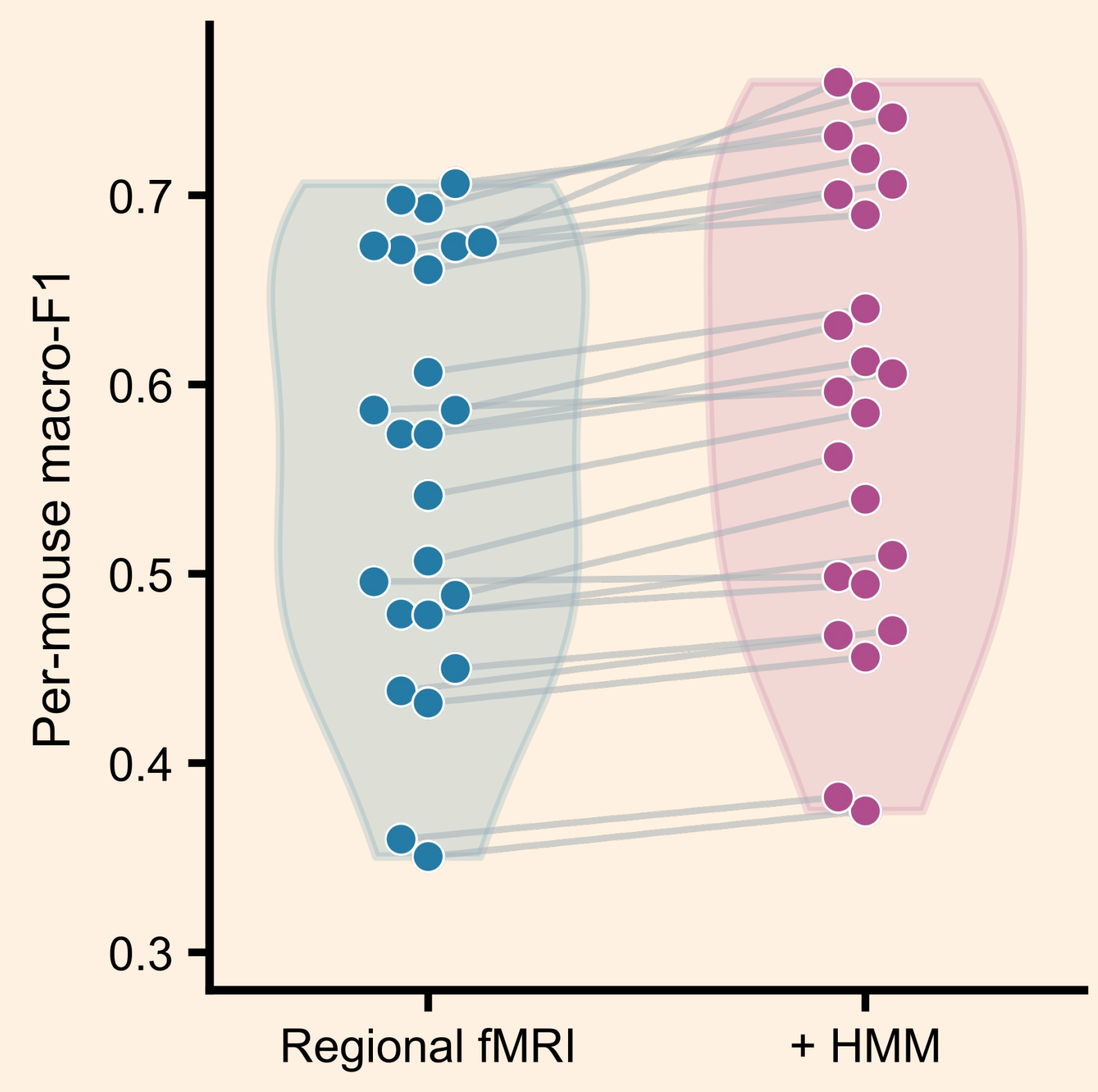


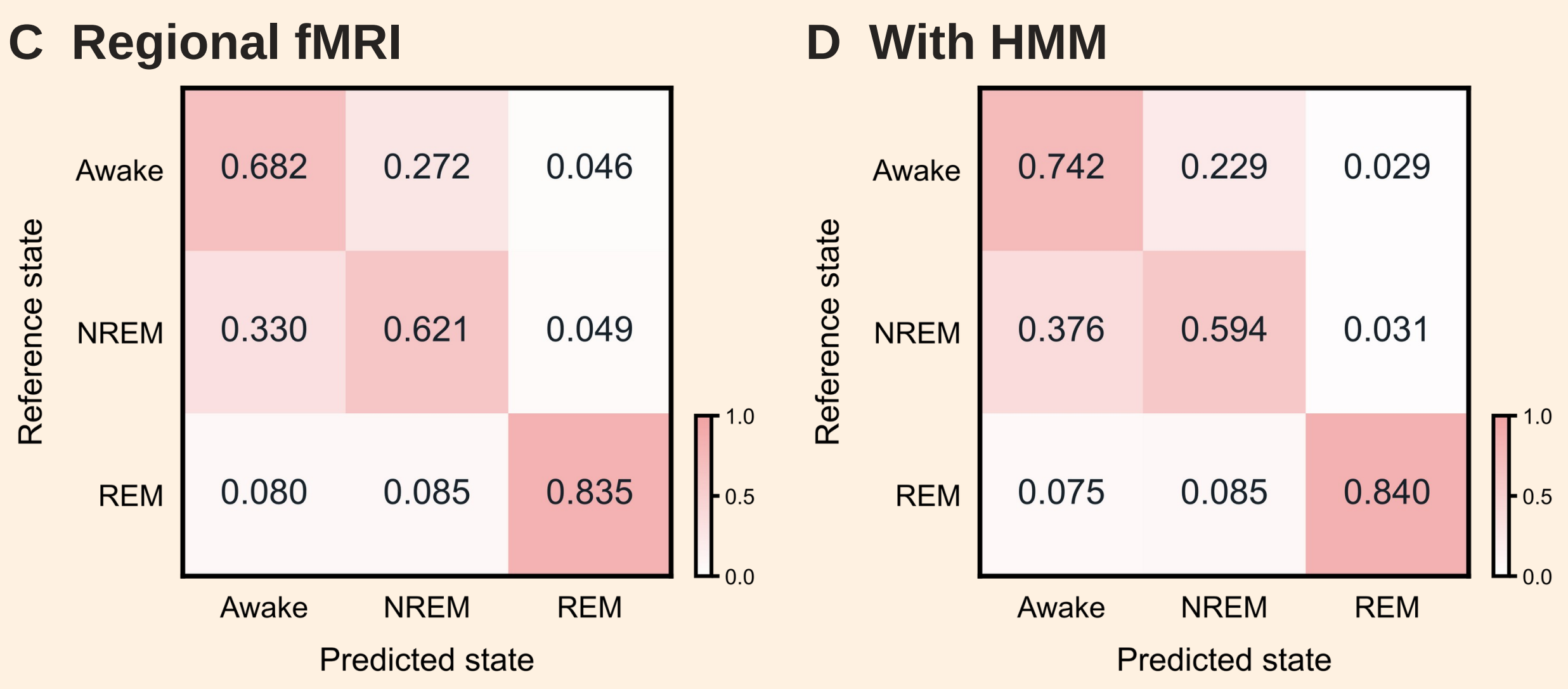
Fig. 1A. Regional features and temporal decoding.

RESULTS

Temporal decoding increased pooled macro-F1 from 0.605 to 0.651. All 24 mice improved, with a mean paired increase of 0.034 (95% confidence interval [CI] 0.028–0.042).

B Paired mouse scores





C Regional fMRI

Reference state \ Predicted state	Awake	NREM	REM
Awake	0.682	0.272	0.046
NREM	0.330	0.621	0.049
REM	0.080	0.085	0.835

D With HMM

Reference state \ Predicted state	Awake	NREM	REM
Awake	0.742	0.229	0.029
NREM	0.376	0.594	0.031
REM	0.075	0.085	0.840

Fig. 1B–D. Lines pair mice (n = 24). Macro-F1 averages F1 across states. Matrices show row-normalized proportions. CIs use mouse bootstraps.

E Continuous sleep-state sequence

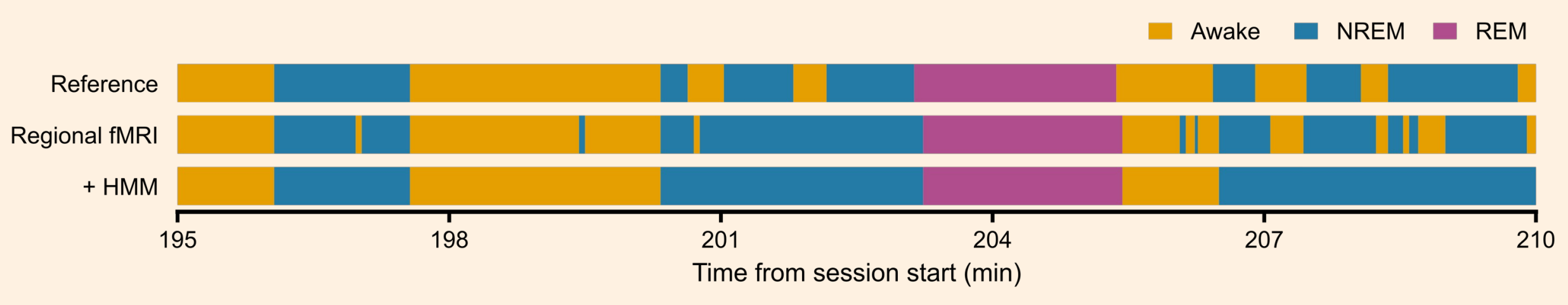


Fig. 1E. Post hoc high-performing 15-min example. REM constituted 3.11% of all volumes.

This gain involved a trade-off: Awake recall rose (0.682 to 0.742), while NREM recall fell (0.621 to 0.594).

RESULTS (CONTINUED)

State-specific regional contributions

A ROI importance

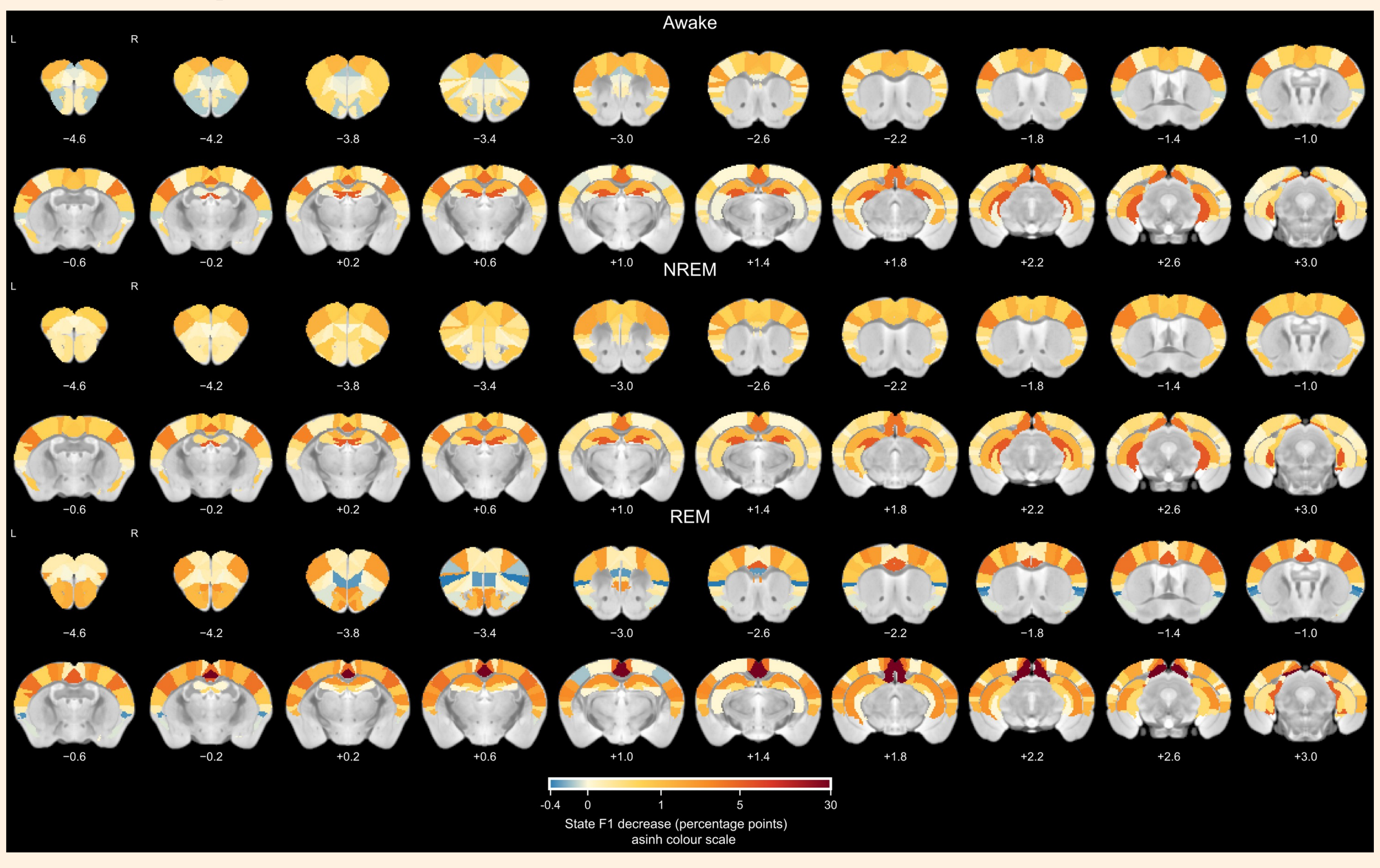


Fig. 2A. Mean held-out F1 loss after 20 joint circular shifts of each ROI's five features. Awake/NREM: 24 mice. REM: 22. Inverse hyperbolic sine (asinh) scale, ticks in percentage points.

B Coefficient contrasts

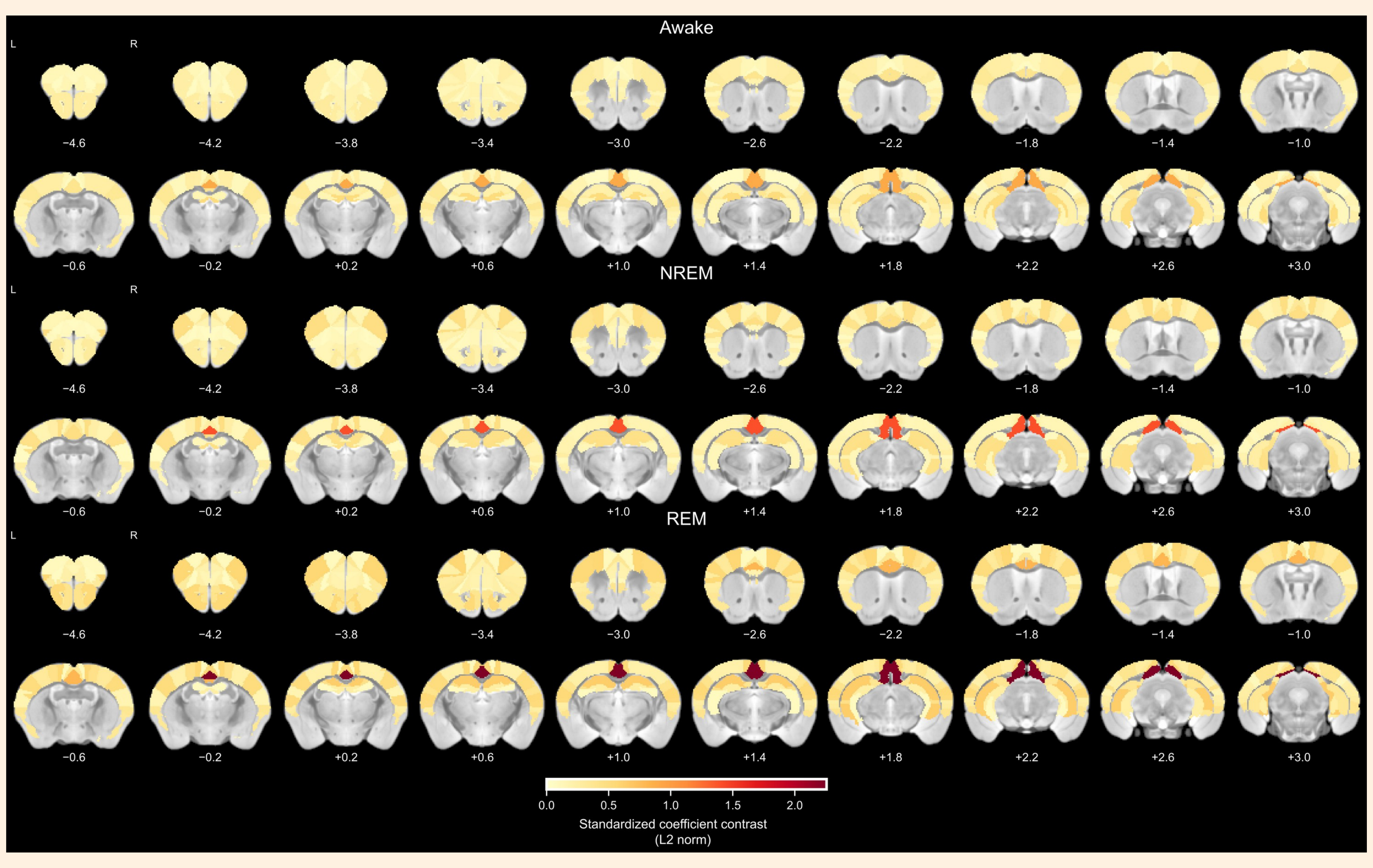


Fig. 2B. Mean Euclidean (L2) norm across five features of standardized coefficient contrasts against the mean of the other two states (24 models, linear scale). Nominal Bregma coordinates in mm.

The dentate gyrus (DG) showed the largest F1 losses for Awake (5.61 percentage points) and NREM (5.69). For REM, ventral retrosplenial cortex (RSPv) ranked first (30.02 points, descriptive 95% CI 24.06–36.49). RSPv also had the largest mean REM coefficient norm.

DISCUSSION

The DG and RSPv findings are consistent with prior reports of NREM-associated infraslow DG activity[3] and retrosplenial activity during REM substage progression[4]. Motion control remained inconclusive: motion-regressed fMRI versus nonlinear motion-only classification gave a paired macro-F1 difference of 0.028 (95% CI −0.001–0.060).

CONCLUSION

Within this dataset, regional fMRI supported continuous sleep-state reconstruction in held-out mice. Independent recordings and further motion controls are needed to assess transfer and use alongside electrophysiological scoring.

REFERENCES

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